

APPROVED AREAS OF FOCUS FOR COMPUTER SCIENCE THESIS

1. General Principles

The Bachelor of Science in Computer Science program focuses on the study, design, development, analysis, and application of computational theories, algorithms, intelligent systems, and emerging computing technologies.

A Computer Science thesis shall primarily contribute to the advancement of computing knowledge through the development, evaluation, enhancement, or application of computational methods, intelligent models, algorithms, architectures, or innovative computing technologies.

Students are encouraged to address real-world problems across domains such as education, healthcare, agriculture, the environment, tourism, disaster risk reduction and management, business, governance, and community development. However, the study's primary contribution must remain within the field of Computer Science.

2. Scope of Acceptable Computer Science Research

To be considered a Computer Science thesis, a proposed study must contain at least one substantial computing component, such as:

- Algorithm development or optimization
- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning
- Natural Language Processing (NLP)
- Computer Vision
- Data Analytics and Predictive Modeling
- Knowledge Representation and Reasoning
- Intelligent Decision Support
- Recommender Systems
- Cybersecurity Mechanisms
- Blockchain Technologies
- Distributed Computing
- Cloud and Edge Computing
- Internet of Things (IoT) Intelligence
- Human-Computer Interaction Research
- Computational Modeling and Simulation
- Computational Science Applications
- Emerging Computing Technologies

The development of a software application serves only as a vehicle for implementing and evaluating the computing contribution.

3. Non-Qualifying Topics

The following projects are generally considered Information Technology projects and are not normally acceptable as Computer Science theses unless enhanced by a significant computing component:

- Student Information Systems
- Enrollment Systems
- Payroll Systems
- Inventory Systems
- Point-of-Sale Systems
- Library Management Systems
- Reservation Systems
- E-Commerce Websites
- General Mobile Applications
- General Database Systems
- School Portals
- Business Information Systems

Example:

Not Acceptable:

"Development of a Web-Based Medical Records System"

Acceptable:

"Development of an Intelligent Medical Records System Using Natural Language Processing for Automated Records Classification and Retrieval"

4. Approved Research Domains

The College adopts three major research domains:

4.1. Machine Learning and Artificial Intelligence

This domain focuses on the development, optimization, evaluation, and application of intelligent algorithms that learn from data, make predictions, generate content, support decision-making, and automate complex tasks.

Possible Areas of Research

A. Artificial Intelligence, Machine Learning, and Generative AI

Description: This area focuses on the development, implementation, evaluation, and optimization of intelligent systems capable of learning, reasoning, prediction, and autonomous decision-making.

Research Areas:

- ➡ Machine Learning
- ➡ Deep Learning
- ➡ Explainable AI
- ➡ Generative AI
- ➡ Large Language Models (LLMs)
- ➡ Retrieval-Augmented Generation (RAG)
- ➡ AI Agents
- ➡ Expert Systems
- ➡ Knowledge-Based Systems
- ➡ Reinforcement Learning
- ➡ Computer Vision
- ➡ Pattern Recognition

Sample Topics

- ✓ AI-Based Student Performance Prediction
- ✓ Deep Learning for Cacao Disease Detection
- ✓ RAG-Based University Policy Assistant
- ✓ Explainable AI for Credit Risk Assessment
- ✓ Intelligent Academic Advising System

B. Data Science, Analytics, and Knowledge Discovery

Description. This domain focuses on extracting meaningful insights and knowledge from data through computational and statistical methods.

Research Areas

- ➡ Data Mining
- ➡ Predictive Analytics
- ➡ Learning Analytics
- ➡ Big Data Analytics
- ➡ Data Visualization
- ➡ Knowledge Discovery
- ➡ Business Intelligence
- ➡ Social Network Analysis

Sample Topics

- ✓ Predictive Analytics for Student Retention
- ✓ Disaster Risk Forecasting Using Machine Learning
- ✓ Tourism Demand Prediction Models
- ✓ Educational Data Mining for Learning Outcomes

C. Intelligent Software Engineering and Computing Systems

Description. This domain focuses on software systems that incorporate advanced computational, analytical, adaptive, or intelligent capabilities.

Research Areas

- ➡ Intelligent Software Systems
- ➡ Adaptive Systems
- ➡ Recommender Systems
- ➡ Software Analytics
- ➡ Automated Testing
- ➡ AI-Assisted Software Engineering
- ➡ Autonomous Software Agents
- ➡ Decision Support Systems

Sample Topics

- ✓ AI-Assisted Code Review System

- ✓ Adaptive E-Learning Platform
- ✓ Intelligent Research Recommendation System
- ✓ Automated Software Defect Prediction Model

4.2. Human-Centered Computing

This domain focuses on designing, developing, and evaluating computing technologies that enhance human interaction, communication, accessibility, productivity, learning, and decision-making.

Possible Areas of Research

A. Natural Language Processing and Conversational AI

Description. This domain focuses on enabling computers to understand, process, generate, and interact using human language.

Research Areas

- | | |
|------------------------------|------------------------|
| ➡ Chatbots | ➡ Sentiment Analysis |
| ➡ Question Answering Systems | ➡ Multilingual Systems |
| ➡ Information Retrieval | ➡ Speech Recognition |
| ➡ Text Mining | ➡ Language Translation |

Sample Topics

- ✓ Multilingual University Chatbot
- ✓ NLP-Based Records Retrieval System
- ✓ Filipino Language Question-Answering System
- ✓ AI-Powered Student Help Desk

B. Human-Computer Interaction and User Experience

Description. This domain focuses on improving interaction between humans and computing technologies.

Research Areas

- | | |
|---------------------------|--------------------------|
| ➡ Human-AI Interaction | ➡ User Experience Design |
| ➡ Accessibility Computing | ➡ Adaptive Interfaces |
| ➡ Assistive Technologies | ➡ Multimodal Interaction |

Sample Topics

- ✓ Accessibility Evaluation of E-Government Systems
- ✓ AI-Based Interface Personalization
- ✓ Voice-Controlled Educational Assistant
- ✓ Adaptive User Interfaces for Learning Systems

C. Game Computing and Serious Games

Description. This domain focuses on computational methods, artificial intelligence, simulation, and interactive technologies in games.

Research Areas

- | | |
|---------------------------------|-----------------------|
| ➡ Game AI | ➡ Serious Games |
| ➡ Procedural Content Generation | ➡ VR/AR Games |
| ➡ Educational Games | ➡ Multiplayer Systems |

Sample Topics

- ✓ AI-Based NPC Behavior System
- ✓ Educational Programming Game
- ✓ Serious Game for Disaster Preparedness
- ✓ Procedural Generation of Learning Scenarios

4.3. Artificial Perception

This domain focuses on enabling computers to perceive, interpret, understand, and interact with the physical and digital environment using visual, auditory, sensor, and contextual information.

Possible Areas of Research

A. Computer Vision, Graphics, AR, VR, and Extended Reality

Description. This area focuses on visual computing and immersive technologies.

Research Areas

- | | |
|----------------------|---------------------|
| ➡ Computer Vision | ➡ Augmented Reality |
| ➡ Image Processing | ➡ Virtual Reality |
| ➡ Object Detection | ➡ Mixed Reality |
| ➡ Facial Recognition | ➡ Digital Twins |

Sample Topics

- ✓ AR-Based Tourism Guide
- ✓ Deep Learning for Traffic Monitoring
- ✓ VR Disaster Preparedness Simulator
- ✓ Digital Twin for Smart Campus Management

B. Internet of Things (IoT), Embedded Systems, and Smart Technologies

Description. This area focuses on intelligent sensing, monitoring, and automation through interconnected devices.

Research Areas

- | | |
|--------------------------------|--------------------|
| * Smart Agriculture | * Smart Healthcare |
| * Smart Environment Monitoring | * Smart Cities |

* Intelligent Embedded Systems

* Real-Time Systems

* Sensor Networks

Sample Topics

- ✓ IoT-Based Flood Monitoring with Predictive Analytics
- ✓ Smart Irrigation Using Machine Learning
- ✓ Intelligent Energy Consumption Monitoring
- ✓ Edge AI for Environmental Monitoring

C. Computational Science and Interdisciplinary Computing

Description. This domain focuses on applying computational techniques to solve problems in other disciplines.

Research Areas

- | | |
|-----------------------------|--------------------------------|
| ➡ Bioinformatics | ➡ Environmental Informatics |
| ➡ Health Informatics | ➡ Disaster Informatics |
| ➡ Computational Agriculture | ➡ Computational Social Science |

Sample Topics

- ✓ AI-Based Disease Surveillance System
- ✓ Computational Analysis of Genetic Data
- ✓ Disaster Risk Prediction Models
- ✓ Cacao Pest Detection Using Deep Learning

4.4 Cross-Cutting Application Domains

These represent societal and industry application areas where computing technologies from the three research domains may be applied. Students are encouraged to align their research with these priority areas while demonstrating a clear contribution to computing.

Possible Areas of Research

A. Sustainable Computing and Smart Communities

Description. This domain focuses on leveraging computing technologies to address societal and sustainability challenges.

Research Areas

- | | |
|--|--------------------------------------|
| ➡ Disaster Risk Reduction and Management | ➡ Smart Communities |
| ➡ Climate Informatics | ➡ Community Decision Support Systems |
| ➡ Sustainable Development Analytics | |

Sample Topics

- ✓ Smart Community Information Platform
- ✓ Climate Risk Analytics Dashboard

- ✓ Microenterprise Resilience Monitoring System
- ✓ Intelligent Disaster Response Decision Support System

B. Cybersecurity and Digital Forensics

Description. This domain focuses on protecting information systems and digital assets against threats and attacks.

Research Areas

- | | |
|-----------------------|---------------------------------|
| ➡ Intrusion Detection | ➡ Vulnerability Assessment |
| ➡ Threat Intelligence | ➡ Privacy-Preserving Analytics |
| ➡ Malware Analysis | ➡ Blockchain Security |
| ➡ Digital Forensics | ➡ Secure Authentication Systems |

Sample Topics

- ✓ Machine Learning-Based Intrusion Detection System
- ✓ Blockchain-Based Identity Verification
- ✓ Malware Classification Using Deep Learning
- ✓ Digital Forensics Investigation Framework

C. Blockchain and Emerging Computing Technologies

Description. This domain focuses on next-generation computing paradigms and decentralized technologies.

Research Areas

- | | |
|---------------------------|------------------------------|
| ➡ Blockchain Applications | ➡ Decentralized Applications |
| ➡ Smart Contracts | ➡ Web3 Technologies |
| ➡ Digital Identity | |

Sample Topics

- ✓ Blockchain-Based Academic Credential Verification
- ✓ Smart Contract-Based Procurement System
- ✓ Decentralized Records Repository
- ✓ Digital Identity Management Platform

5. Alignment with the United Nations Sustainable Development Goals (SDGs)

In selecting a thesis topic, students are strongly encouraged to consider the potential contribution of their proposed study to the achievement of the United Nations Sustainable Development Goals (SDGs). The SDGs provide a global framework for addressing critical societal challenges, including poverty, health, education, economic growth, environmental sustainability, climate action, and resilient communities.

As a discipline, Computer Science plays a significant role in developing innovative computing solutions that support evidence-based decision-making, intelligent automation, data-

driven governance, digital transformation, and sustainable development. Students are therefore encouraged to identify real-world problems whose solutions may contribute to one or more SDGs while maintaining a substantial computing or computational contribution.

The alignment of a thesis topic with the SDGs strengthens its societal relevance, research significance, and potential impact on communities, industries, government agencies, and other stakeholders. Such alignment may also increase opportunities for research dissemination, funding support, interdisciplinary collaboration, and publication in reputable journals and conferences. While SDG alignment is encouraged, the primary contribution of the study must remain within the field of Computer Science through the development, evaluation, enhancement, or application of computational methods, intelligent systems, algorithms, models, or emerging computing technologies.

Examples of Computer Science Thesis Topics Aligned with the Sustainable Development Goals

SDG	Description	Example Computer Science Thesis Topics
SDG 2: Zero Hunger	Improve food security and sustainable agriculture	Deep Learning-Based Cacao Disease Detection System
SDG 3: Good Health and Well-Being	Improve healthcare services and health outcomes	AI-Assisted Disease Surveillance and Prediction System
SDG 4: Quality Education	Enhance access to quality education	Intelligent Learning Analytics System for At-Risk Students
SDG 8: Decent Work and Economic Growth	Promote sustainable economic growth and productivity	Predictive Analytics for Microenterprise Resilience Assessment
SDG 9: Industry, Innovation and Infrastructure	Foster innovation and technological advancement	Edge AI-Based Smart Manufacturing Monitoring System
SDG 11: Sustainable Cities and Communities	Create safer and more sustainable communities	Smart Flood Monitoring and Early Warning System
SDG 13: Climate Action	Address climate change and disaster resilience	Machine Learning-Based Climate Risk Prediction Model
SDG 16: Peace, Justice and Strong Institutions	Improve governance, transparency, and security	Blockchain-Based Digital Records Verification System
SDG 17: Partnerships for the Goals	Promote collaboration and data sharing	Multi-Stakeholder Disaster Information Management Platform

Topic Selection Guideline: Students are encouraged to identify societal, organizational, environmental, or industrial problems that may benefit from computing-based solutions. Whenever feasible, proposed studies should demonstrate potential contributions to one or more Sustainable Development Goals (SDGs). However, alignment with an SDG alone does not qualify a topic as a Computer Science thesis; the study must still contain a substantial computing

contribution such as algorithm development, artificial intelligence, machine learning, data analytics, cybersecurity, intelligent systems, computational modeling, or other recognized Computer Science domains.

For the specific context in Philippine universities, the most relevant SDGs that commonly align with Computer Science theses are:

- **SDG 3 – Good Health and Well-Being**
- **SDG 4 – Quality Education**
- **SDG 8 – Decent Work and Economic Growth**
- **SDG 9 – Industry, Innovation and Infrastructure**
- **SDG 11 – Sustainable Cities and Communities**
- **SDG 13 – Climate Action**
- **SDG 16 – Peace, Justice and Strong Institutions**
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6. Alignment with the Bicol University Research and Development Agenda

In support of the University's mandate to generate knowledge, foster innovation, and contribute to regional and national development, students are encouraged to align their thesis topics with the priority areas identified in the Bicol University Research and Development Agenda.

The University recognizes the role of research in addressing societal challenges and promoting sustainable development through science, technology, innovation, and evidence-based decision-making. As future computing professionals, Computer Science students are encouraged to develop computing solutions, intelligent systems, analytical models, and emerging technologies that contribute to the achievement of these institutional priorities.

Alignment with the Bicol University Research and Development Agenda enhances the relevance, applicability, and potential impact of a thesis study. It also provides opportunities for interdisciplinary collaboration, external funding, community engagement, technology transfer, and research publication.

Students should identify one or more research agenda categories that are relevant to their proposed study. However, alignment with the University Research Agenda shall not replace the requirement that the thesis must demonstrate a substantial Computer Science contribution.

Bicol University Research and Development Agenda Categories

1. Food Security, Safety, and Poverty Reduction. This thematic area focuses on improving food availability, food safety, agricultural productivity, livelihood opportunities, and poverty alleviation through innovative and sustainable solutions.

Possible Computer Science Applications:

- Artificial Intelligence for crop disease detection
- Precision agriculture using IoT and machine learning

- Agricultural analytics and forecasting
- Food supply chain optimization
- Smart farming technologies
- Decision support systems for farmers

2. Climate Change, Environmental Sustainability, and Natural Resource Management. This thematic area focuses on climate resilience, environmental protection, sustainable resource utilization, biodiversity conservation, and disaster risk reduction.

Possible Computer Science Applications:

- Climate analytics and forecasting
- Environmental monitoring systems
- Disaster prediction models
- Remote sensing and geospatial analytics
- Computer vision for biodiversity monitoring
- Smart environmental sensing networks

3. Global Competitiveness of Business and Industry. This thematic area focuses on innovation, productivity enhancement, digital transformation, competitiveness, and sustainable economic growth.

Possible Computer Science Applications:

- Business intelligence and analytics
- Intelligent recommendation systems
- Process optimization
- Predictive analytics
- Industry 4.0 technologies
- Smart manufacturing systems

4. Social Development Promotion. This thematic area focuses on improving the quality of life through innovations in culture, health, governance, education, and community development.

Possible Computer Science Applications:

Health

- Disease prediction systems
- Health analytics
- Healthcare decision support systems

Education

- Learning analytics
- Intelligent tutoring systems
- Adaptive learning platforms

Governance

- Decision support systems
- Public service analytics

- E-governance technologies

Culture

- Digital heritage preservation
- Augmented reality cultural applications
- Knowledge repositories

5. Institutional Development and Policy Innovation. This thematic area focuses on improving organizational effectiveness, governance, decision-making, policy development, and service delivery within educational institutions, including Bicol University.

Possible Computer Science Applications:

- Policy analytics
- Institutional decision support systems
- Educational data mining
- Knowledge management systems
- Process automation
- Intelligent campus systems

For proposal evaluation, I would require every title proposal to explicitly indicate:

Criterion	Example
CS Domain	Artificial Intelligence
Computing Contribution	Deep Learning Classification Model
BU Agenda	Food Security, Safety, and Poverty Reduction
SDG	SDG 2: Zero Hunger
Beneficiary	Cacao Farmers
Expected Innovation	Automated disease detection with explainable AI